Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 1.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. The length of each side of square A is increased by 100 percent to make square B. If the length of the side of square B is increased by 50 percent to make square C, by what percent is the area of square C greater than the sum of the areas of squares A and B?

(A) 75% (B) 80% (C) 100% (D)150% (E) 180%

2. If $a = 6 - b^3$ and $b = -2$, what is the value of a?

(A) -8 (B) -2 (C) 2 (D) 14 (E) 18

3. For all z, $3^z + 3^z + 3^z =$

(A) $3^z + 3$ (B)3^{z+1} (C) 3^{z+3} (D) 3^{3z} (E) 9^z

4. For all $c \neq \pm\frac{1}{5}$, $\frac{5c^2-24c-5}{1-25c^2} =$

(A) $\frac{5c+1}{5+c}$ (B)$\frac{5+c}{5c+1}$ (C) $-\frac{5c-1}{5-c}$ (D) $-\frac{5-c}{5c-1}$ (E) $\frac{5-c}{5c-1}$

5. The only value of a for which the simultaneous equations $\begin{cases} 2x + 3y = 5 \\ x + ay = 2 \end{cases}$ would have no solution is

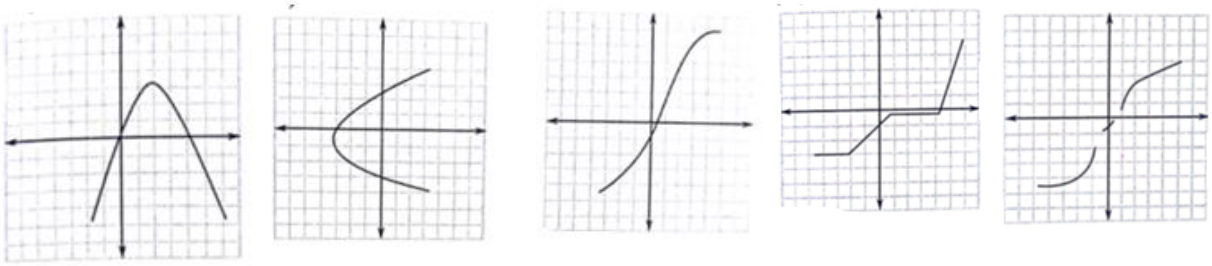
(A) 2 (B) 0 (C) 1.5 (D) 0.5 (E) -0.5

6. $\sqrt{9 + 4\sqrt{2}}$=

(A)$3 + 2\sqrt{2}$ (B) $3 + 2\sqrt[4]{2}$ (C) $2 + 3\sqrt{2}$ (D) $2 + 3\sqrt[4]{2}$ (E) $1 + 2\sqrt{2}$

7. Which of the following graphs is not a function?

(A) (B) (C) (D) (E)



8. A parallelogram is inscribed in a rectangle such that its vertices are the midpoints of the sides of the rectangle. If the area of the parallelogram is 60, and one of the dimensions of the rectangle is 12, what is the other dimension of the rectangle?

(A) 8 (B) 10 (C) 12 (D) 14 (E) 16

9. If $f(x) = 2x^3$ and $g(x) = 3x$, what is the value of $g(f(-2)) - f(g(-2))$?

(A) -480 (B) -384 (C) 0 (D) 384 (E) 480

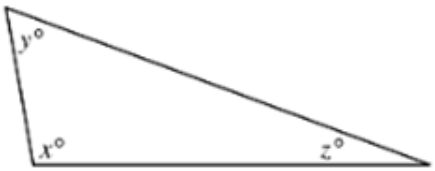
10. Assume that b and c are integers greater than one. In base b, c^2 is written as 10. Then b^2, when written in base c, is

(A) 100 (B) 1001 (C) 101 (D) 1010 (E) 10000

11. For a certain integer n, $5n + 16$ and $8n + 29$ have a common factor larger than one. That common factor is

(A) 11 (B) 13 (C) 17 (D) 19 (E) 23

12. In the figure below, $x > 90$ and $y = z + 1$. If z is an integer, what is the greatest possible value of y?



(A) 30 (B) 45 (C) 60 (D) 61 (E) 89

13. If the first term in a geometric sequence is 3, and if the third term is 48, what is the $11th$ term?

(A) 228 (B) 528 (C) 110,592 (D) 3,145,728 (E) 12,582,912

14. Two fair dice are tossed simultaneously. What is the probability that the product of the numbers that land showing is at least 25?

(A) $\frac{1}{36}$ (B) $\frac{1}{12}$ (C) $\frac{1}{9}$ (D) $\frac{1}{4}$ (E) $\frac{3}{2}$

15. Ten points lie on the circumference of a circle. Which of the following is the value that results when the number of triangles that can be created by connecting these points is subtracted from the number of heptagons (seven-sided polygons) that can be created by connected these points?

(A) 210 (B) 35 (C) 21 (D) 4 (E) 0

16. If $f(t) = t2 + \frac{3}{2}t$, then $f(q-1) =$

(A) $q^2 - \frac{3}{2}$ (B) $q^2 - \frac{1}{2}q$ (C) $q^2 - \frac{1}{2}q + \frac{1}{2}$ (D) $q^2 - \frac{1}{2}q - \frac{1}{2}$ (E) $q^2 - \frac{1}{2}q + 1$

17. How many integers are in the solution set of $|2x + 6| < \frac{19}{2}$?

(A) None (B) Two (C) Nine (D) Fourteen (E) Infinitely many

18. If $f(x) = \frac{\sqrt{x^2-4}}{x-4}$, what are all the values of x for which $f(x)$ is defined?

(A) All real numbers except 4

(B) All real numbers except -2 and 2

(C) All real numbers greater than or equal to -2 and less than or equal to 2

(D) All real numbers less than or equal to -2 or greater than or equal to 2

(E) All real numbers less than or equal to -2 or greater than or equal to 2, except 4

19. On the second leg of a certain journey, a person traveled at an average speed 50 percent greater than the average speed at which the person traveled the first leg of the journey, and for an amount of time 50 percent greater than the amount of time spent on the first leg of the journey. The person's average speed for the entire journey was that percent greater than the person's average speed on the first leg of the journey?

(A) 22.5% (B) 25% (C) 30% (D) 50% (E) 100%

20. Last year a manufacturer had a sales total for January that was 50 percent greater than the average (arithmetic mean) of the monthly sales totals for February through December. The sales total for January was that fraction of the sales total for the year?

(A) $\frac{1}{11}$ (B) $\frac{1}{8}$ (C) $\frac{3}{25}$ (D) $\frac{3}{22}$ (E) $\frac{3}{5}$

21. For all nonzero numbers x and y, the operation ∇ is defined by the equation $x \nabla y = \frac{|x|}{x^2} - \frac{y^2}{|y|}$ when $x > y$ and by the equation $x \nabla y = \frac{|x|}{y^2} - \frac{x^2}{|y|}$ when $x \leq y$. If $x \nabla y < 0$, then which of the following could also be true?

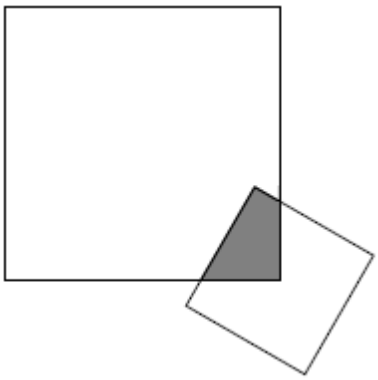
I. $x^3 = y^3$

II. $(y + x)(y - x) > 0$

III. $x - y > 0$

(A) I only (B) II only (C) III only (D) I and II (E) II and III

22. A 5×5 square card is laid on a 10×10 square card so that the center of the small card lies directly over the corner of the large card. The area of the covered region is



(A) $10\frac{\sqrt{2}}{3}$ (B) $\frac{75}{14}$ (C) $\frac{25}{4}$ (D) $3(1 + \sqrt{2})$ (E) 6

23. In $figure\ 1$, the lateral area of the right circular cone is 60π. If the radius of the base is 6, what is the volume of the cone?

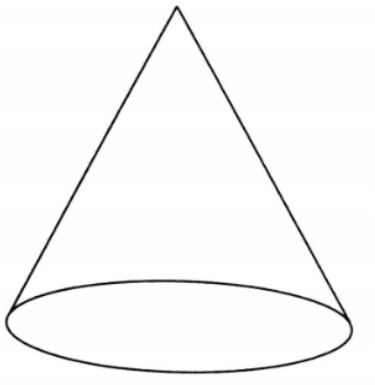


Figure 1

(A) 96π (B) 108π (C) 120π (D) 184π (E) 288π

24. In $Figure\ 4$, the perimeter of isosceles trapezoid $WXYZ$ is 62. If $ZY = 7$ and $WX = 25$, what is the length of diagonal WY?

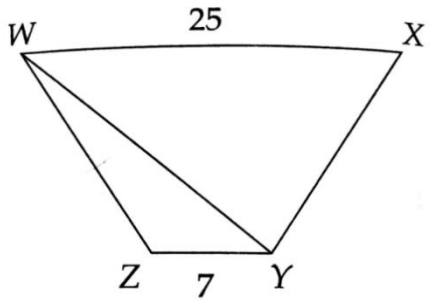


Figure 4

(A) 15 (B) 17 (C) 20 (D) 25 (E) 26

25. In the rectangular solid in $Figure\ 1$, what is the distance from vertex R to vertex T?

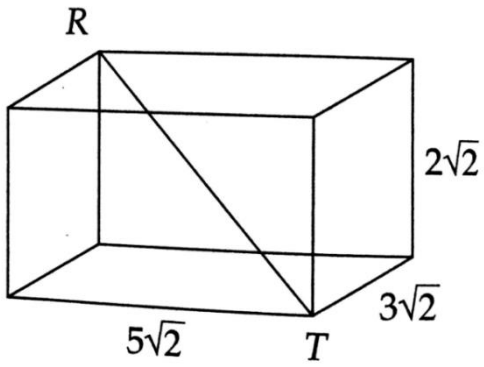


Figure 1

(A) $2\sqrt{19}$ (B) $2\sqrt{17}$ (C) $8\sqrt{2}$ (D) $10\sqrt{2}$ (E) 30

26. In $figure\ 6$, points P, Q and R lie on the circumference of the circle centered at O. If angle OPQ measures $x°$ and angle QRO measures $y°$, what is the measure of angle POR in terms of x and y?

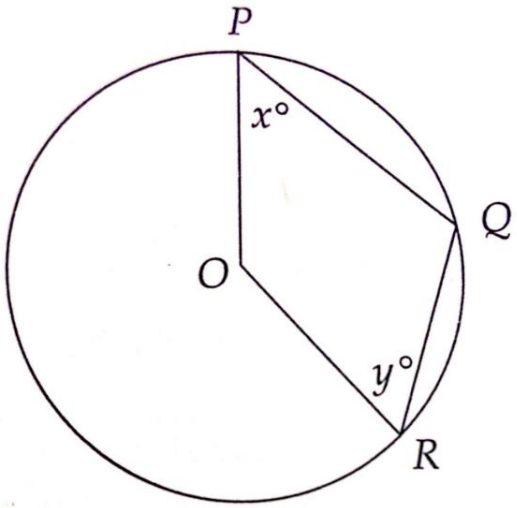


Figure 6

(A) $(360 + x + y)°$

(B) $(360 - x - y)°$

(C) $(360 - 2x - 2y)°$

(D) $(180 + x + y)°$

(E) $(180 - 2x - 2y)°$

27. A delicatessen charges for a certain type of salad at a rate of 2 dollars for the first half pound and 75 cents for each additional half pound. Which of the following expressions represents the total charge, in cents, for p pounds of the salad, where p is a positive integer?

(A) $25(6p + 8)$ (B) $25(6p + 5)$ (C) $25(3p + 5)$ (D) $1.25 + 1.5p$ (E) $2 + 0.75(p - 1)$

28. Which of the following is a point at which the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$ intersects the y-axis?

(A) $(0, -3)$ (B) $(0, -4)$ (C) $(0, -8)$ (D) $(0, -9)$ (E) $(0, -16)$

29. If $n \neq 3p$, and $s = \frac{n^2+p}{n-3p}$, what is the value of p in terms of n and s?

(A) $\frac{n^2-ns}{1-3s}$ (B) $\frac{ns+n^2}{3s+1}$ (C) $-\frac{ns-n^2}{3s+1}$ (D) $\frac{n^2-ns}{3s+1}$ (E) $-\frac{n^2-ns}{3s+1}$

30. Which of the following equations can be a solution to the pair of equation $2q - rt = 21$ and $q + rt = 36$?

(A) $q = 18$ and $t = 2$ (B) $q = 19$ and $t = 1$ (C) $q = 20$ and $t = -1$ (D) $q = 21$ and $t = -2$ (E) $q = 23$ and $t = -3$

Answer key:

№	Correct Answer	Your answer
1	B	
2	D	
3	B	
4	E	
5	C	
6	E	
7	B	
8	B	
9	D	
10	E	
11	C	
12	B	
13	D	
14	C	
15	E	
16	D	
17	C	
18	E	
19	C	
20	C	
21	D	
22	C	
23	A	
24	C	
25	A	
26	C	
27	B	
28	B	
29	E	
30	B	
		Total: _____